RPS Research

October 2025

Price Targets
CAD 1.20

Defense Metals Corp

DEFN.V
Sector: Mining

Defense Metals is gearing up to turn rare earth dirt into rare earth dollars

Executive Summary

Defense Metals represents a **high-risk, high-reward development play** on Western rare earth supply chain diversification, with Wicheeda positioned as the **most advanced undeveloped NdPr-focused project in North America**. The February 2025 PFS demonstrates compelling economics (**US $1.8B pre-tax NPV, 24.6% IRR**) underpinned by an **exceptional 87.3% NdPr product basket** and **proven mineral reserves of 25.5 million tonnes at 2.43% TREO**. Recent balance sheet strengthening (CAD $15M financing, debt elimination, EDC US $250M Letter of Interest) de-risks near-term execution, though the company faces 4-6 year timeline to production and requires approximately CAD $1 billion in total project financing. We assign a **SPECULATIVE BUY** rating with CAD $1.20 price target based on 60% probability-adjusted NAV, reflecting binary development risk, shareholder dilution, and current NdPr pricing significantly below the 2022 high of US $170-180/kg vs. US $70-80/kg, especially considering the current political climate with trade restriction uncertainty always mounting over the rare earth industry as China controls majority of the world's rare earth assets.

Key Highlights

Robust PFS Economics: Pre-tax NPV of US $1.8 billion (8% discount), IRR 24.6%, after-tax NPV US $992M, IRR 19.2%, 3.7-year payback on US $1.4B initial capex for 15-year mine life

Best-in-Class Product Mix: 87.3% NdPr concentration in final MREC product—among highest globally—targeting premium permanent magnet feedstock markets with operating cash breakeven US $37.42/kg NdPrO

Proven Reserves Status: Only North American rare earth project not yet in production with proven mineral reserves (25.5M tonnes at 2.43% TREO), 50% TREO concentrate at 81% recovery via simple flotation

Balance Sheet Strengthened: CAD $15M October 2025 financing, CAD $4M debt conversion, debt-free with positive working capital; EDC Letter of Interest for up to US $250M as Mandated Lead Arranger

Development Timeline: DFS commencement H1 2026, environmental assessment submission 2027, permits 2028-2029, first production 2029-2030 contingent on financing and permitting success

Key Risks: NdPr pricing should sustain the project breakeven US $67.60/kg; CAD $1B financing requirement implies 50-100% shareholder dilution; 3-7 year BC permitting timelines; no binding offtake agreements

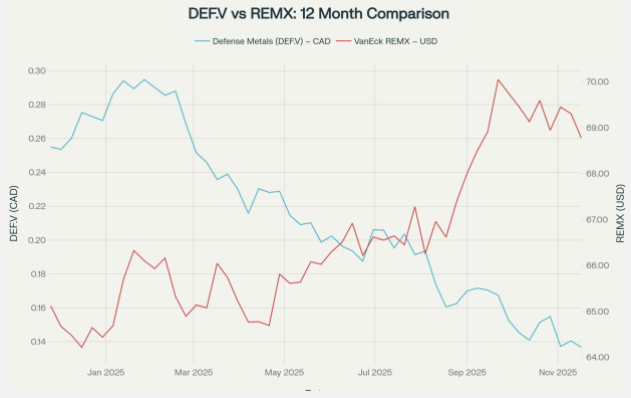
Industry

The global rare earth elements industry remains highly concentrated, with China controlling 69% of annual mine production, 85% of refining and processing capacity, and nearly 90% of the world's permanent magnet manufacturing—a critical segment for EVs, wind turbines, and defense technologies. This dominance is both a source of geopolitical leverage and supply chain risk, as China's export restrictions, price controls, and quotas can ripple through downstream industries globally. North America has only a handful of active or near-term rare earth projects, with MP Materials' Mountain Pass (California) currently the sole large-scale producer, and emerging players like Defense Metals, American Rare Earths, USA Rare Earth, NioCorp, and Ucore advancing different stages of mining or processing assets. Strategic government funding support, permit reform, and offtake agreements are accelerating domestic supply chain builds, but until more projects reach commercial scale, Western supply remains highly vulnerable to

About the Company

Defense Metals Corp. (TSXV: DEFN.V, OTCQB: DFMTF, FSE: 35D) is a Canadian junior mining company focused exclusively on developing its 100%-owned Wicheeda rare earth element project located 80 kilometers northeast of Prince George, British Columbia. Founded in 2017, the company employs a development strategy targeting high-purity, NdPr-dominated rare earth deposits with simple beneficiation characteristics to minimize processing costs and maximize product value. Defense Metals' technical team has advanced Wicheeda from grassroots exploration through Pre-Feasibility Study, leveraging expertise in flotation metallurgy and hydrometallurgical processing to establish the asset as a Western-aligned, non-Chinese rare earth supply solution for North American battery and permanent magnet manufacturing supply chains. Defense Metals represents a development-stage play on critical minerals supply chain diversification and geopolitical reshoring trends, though the company faces multi-year execution risk, significant financing requirements, and commodity price sensitivity before achieving first production targeted for 2029-2030.

Metric	Unit/Notes	Value
Mineral Reserves and Resources		
Proven and Probable Reserves - Tonnage	Million Tonnes	25.5
Proven and Probable Reserves - Average Grade	% TREO	2.43
Measured and Indicated Resources - Tonnage	Million Tonnes	29.2
Measured and Indicated Resources - Average Grade	% TREO	2.27
Inferred Resources - Tonnage	Million Tonnes	5.5
Inferred Resources - Average Grade	% TREO	1.42
Updated Measured + Indicated Resources - Tonnage	Million Tonnes (from company bio)	34.2
Updated Measured + Indicated Resources - Average Grade	% TREO (from company bio)	2.02
Stock and Financial Metrics (October 31, 2025)		
Current Price (Close)	CAD	0.33
52-Week Range	CAD	0.0900 - 0.4350
Volume	Shares	2,576,754
Average Volume	Shares (recent trailing average)	1,051,186
Market Cap	Million CAD (calculated based on close price and estimated shares outstanding of ~391.5 million)	129.204



Investment Thesis

Exceptional Deposit Quality and Product Mix: Wicheeda's investment case is anchored by its world-class technical profile: 29.2 million tonnes of measured and indicated resources at 2.27% TREO, with proven and probable reserves of 25.5 million tonnes at 2.43% TREO. The deposit's monazite-dominated mineralogy enables simple flotation beneficiation delivering a 50% TREO concentrate at 81% recovery—placing Wicheeda in an elite class alongside Lynas' Mt. Weld (Australia) and MP Materials' Mountain Pass (California) as one of the few globally competitive rare earth deposits requiring no complex acid pre-treatment. This metallurgical simplicity translates directly into lower processing capex and operating costs relative to competing projects. More importantly, Wicheeda's final mixed rare earth carbonate (MREC) product contains 87.3% neodymium-praseodymium (NdPr) by weight of total rare earth oxide content—among the highest NdPr concentrations of any rare earth project globally. This NdPr-heavy basket is achieved through solvent extraction removal of low-value cerium and lanthanum, eliminating exposure to oversupplied commodity rare earth markets (where cerium trades at ~$2-3/kg) and concentrating revenue on the highest-value magnetic rare earths used in permanent magnets for electric vehicles, wind turbines, and defense applications (where NdPr trades at $50-75/kg, with historical peaks above $150/kg). The company achieved a metallurgical breakthrough in January 2025, demonstrating 85% NdPr purity in bench-scale testing, further validating commercial product specifications and de-risking the flowsheet for DFS.

Strategic Positioning in Western Supply Chain Diversification: Defense Metals benefits from powerful geopolitical tailwinds as Western governments seek to diversify critical minerals supply chains away from Chinese dominance. China currently controls approximately 85% of global rare earth processing capacity and 90% of permanent magnet production, creating strategic vulnerabilities for North American and European automotive, renewable energy, and defense industries. The U.S. Department of Defense, Department of Energy, and Export-Import Bank have allocated billions in funding for non-Chinese rare earth supply development, while Canada's Critical Minerals Strategy designates rare earths as Tier 1 strategic materials eligible for accelerated permitting and financial support. Wicheeda's location in British Columbia—a Tier 1 mining jurisdiction with established rule of law, indigenous consultation frameworks, and existing infrastructure (paved highway access, hydropower, natural gas pipelines)—positions the project to capture premium pricing for "non-Chinese" rare earth material. Management reports ongoing offtake discussions with Western end-users and has executed a non-binding MOU for a significant supply agreement, though commercial terms remain undisclosed. The project's 87.3% NdPr basket is specifically designed to serve permanent magnet manufacturers seeking secure, traceable feedstock, potentially commanding 10-20% premiums over Chinese-origin material as ESG and supply chain traceability requirements intensify.

Credible Path to Production with Institutional Support: Defense Metals has made material progress de-risking its pathway to production over the past 12 months. The February 2025 PFS established proven reserves status—making Wicheeda the only North American rare earth project not yet in production with proven reserves—and demonstrated robust economics with after-tax NPV of US $992 million and IRR of 19.2% at base case pricing assumptions. Export Development Canada issued a Letter of Interest in 2025 for up to US $250 million as Mandated Lead Arranger for project financing, representing approximately 18% of the US $1.4 billion initial capex requirement and signaling institutional confidence in project fundamentals. The company has secured indigenous support through a 2024 Joint Co-Design Agreement with the McLeod Lake Indian Band and received public backing from British Columbia's Minister of Mining and Critical Minerals, who committed to improving permitting timelines. In October 2025, Defense Metals eliminated its debt overhang by converting CAD $4 million in convertible notes and raised CAD $15 million in equity financing, achieving a debt-free balance sheet with positive working capital for the first time since advancing to PFS. Management plans to commence a bankable Definitive Feasibility Study in H1 2026, with ongoing metallurgical optimization, pilot plant testing, and environmental baseline work to support EA submission in 2027. While significant risks remain (detailed in Risks section), Defense Metals has established a credible timeline to production decision by 2027-2028, with first production achievable by 2029-2030 if financing and permitting proceed on schedule.

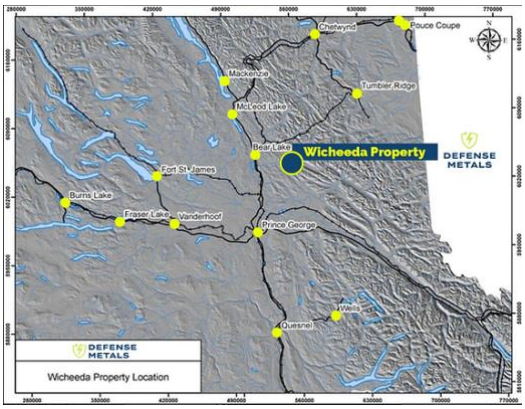
Project Financing Roadmap

The pathway from current position to construction decision requires approximately CAD $1.0-1.2 billion in total capital, structured across three phases:

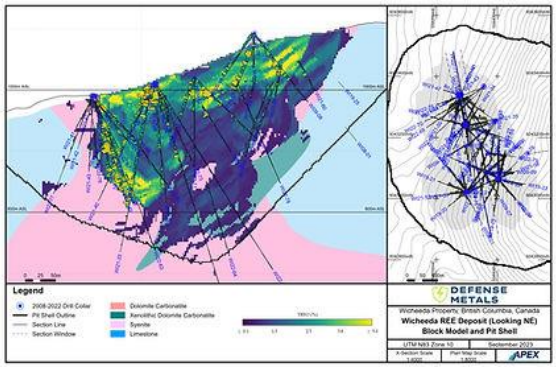
Phase 1 – DFS and Permitting (2026-2028): CAD $25-35M: Funding requirement for DFS completion, pilot plant construction and operation, environmental assessment preparation and submission, indigenous consultation, and early-stage permitting activities. Defense Metals will require 1-2 additional equity raises of CAD $15-20 million to bridge this period, implying 30-50% dilution from current share count at $0.40-0.50 pricing levels.

Phase 2 – Construction Financing (2028-2029): CAD $1.4B: Initial capex of US $1.4 billion (CAD $1.9B at 1.35 FX) for mine development, processing plant construction, and infrastructure. Anticipated financing structure: (1) EDC-led senior debt of US $250-400M (per LOI); (2) strategic equity investment of US $200-300M from mining major or end-user (offtake prepayment); (3) government grants/loans of US $100-200M from Canada Critical Minerals Infrastructure Fund or U.S. Defense Production Act; (4) remaining US $500-700M from equity markets, implying significant dilution if raised at current valuations.

Phase 3 – Working Capital and Ramp-Up (2029-2030): CAD $150-200M: Working capital facility and cost overrun contingency to fund commissioning, ramp-up to commercial production, and initial operating losses during first 12 months of operations.



Wicheeda Mining Site in British Columbia



Wicheeda Long Section - 2023

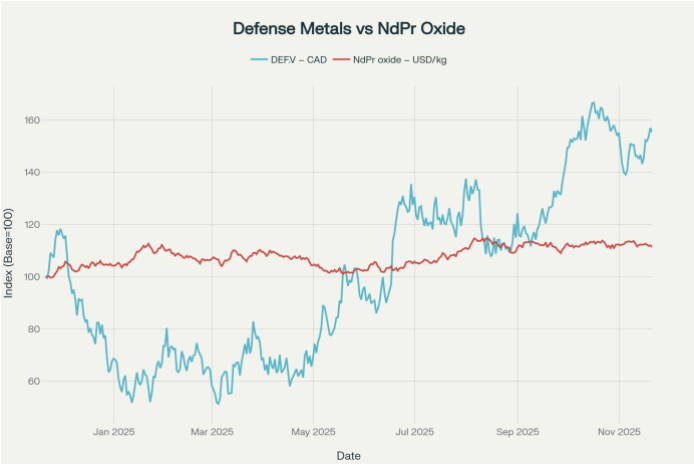
Peer Analysis (Company)

Company	Status	Market Cap (USD M)	EV (USD M)	Price/Book (mrq)	Debt/Equity (mrq)	Current Ratio (mrq)	NdPr Basket	Jurisdiction
Defense Metals	**PFS (Feb 2025) → DFS H1-26**	**72**	**53**	**1.86**	**0.12%**	**1.73**	**87.30%**	**Canada (BC)**
Arafura Rare Earths	Fully financed, construction start 2026	582	564	6.39	0.0016	8.37	~85%	Australia
MP Materials	Producing + Stage 2 separation ramping	10,100	9,160	5.73	0.4351	8.05	70-75%	USA
Lynas Rare Earths	Producing + Malaysia + Kalgoorlie + Texas expansions	9,780	9,880	6.64	0.0859	2.83	65-70%	Australia/Malaysia
Appia Rare Earths & Uranium	Early-stage exploration	30	29	1.63	N/A	2.48	Variable	Canada
Vital Metals	Nechalacho stalled	22	23	0.69	0.0136	0.43	~70%	Canada
Northern Minerals	Heavy REE focus (Dysprosium)	211	187	180.42	9.352	1.17	Heavy REE	Australia

Peer Analysis (Projects)

Project / Company	Jurisdiction	Status	Proven + Probable Reserves (Mt @ %TREO)	NdPr in Basket	Annual NdPr Oxide eq. (tpa)	Initial Capex (US$ bn)	First Production Target
Wicheeda (Defense Metals)	**Canada (BC)**	**PFS (Feb 2025) → DFS H1-26**	**25.5 Mt @ 2.43%**	**87.30%**	**~4,500**	**1.4**	**2029-30**
Mountain Pass (MP Materials)	USA (CA)	Producing (only Western large-scale)	~20 Mt remaining life @ ~6-7%	~70-75%	~3,000–4,000 (2025 actual)	Fully built	Operating
Mt Weld + Malaysia/Texas	Australia + Malaysia	Producing + expansion	>30 Mt @ ~8-9%	~65-70%	~7,200 expanding to 12,000	Fully built + expansions	Operating
Nolans (Arafura)	Australia (NT)	Fully financed, construction imminent	19.6 Mt @ 2.9%	~85%	4,440	~1.2	2027-28
Donald JV (Energy Fuels / Astron)	Australia (VIC)	Permitted, FID late 2025	Large monazite/xenotim	~70% + heavy REE focus	Phase 1: ~1,000 NdPr + heavies	~0.35 (Phase 1)	2027

NdPr Pricing



NdPr pricing information obtained from TradingEconomics

NdPr prices have remained exceptionally firm throughout 2025, reflecting persistent supply constraints and continuous demand momentum from EV and wind sectors, as well as geopolitical initiatives to diversify rare earth supply chains away from China. Recent price strength is also underpinned by slow permitting and development timelines for ex-China projects, plus strategic stockpiling by downstream users as a hedge against potential future disruptions.

Despite the clear fundamental linkage, DEF.V's share price continues to show a relatively low short-term correlation to NdPr spot movements, largely due to its current position as a non-producing, exploration-stage asset. The stock is more responsive to project-specific catalysts and broader risk sentiment than to incremental changes in NdPr pricing; only as DEF.V approaches construction and commercial agreements should its market value begin to more directly reflect NdPr macro trends.

Valuation

Defense Metals' Wicheeda project is valued using a risk-adjusted NAV approach, anchored by its February 2025 Pre-Feasibility Study (PFS). The base case after-tax NPV is US $992M (CAD $1.35B at 1.35 FX), applying a discount rate of 8% and NdPr price assumptions of US $75/kg. Given development-stage risks, including permitting, financing, and dilution, a success probability of 60% is warranted. Adjusting for future equity raises (150M forecast diluted shares post-DFS), our NAV per share is CAD $3.17. For pre-production developers, markets typically apply a 40% discount to risked NAV; thus, our one-year price target for Defense Metals is **CAD $1.20 per share**, offering substantial upside from current trading near $0.45. Key sensitivity: If NdPr prices fall below breakeven ($67.60/kg), NAV could compress towards CAD $1.00 per share; conversely, a pricing rally (to $90/kg+) would support CAD $2.00–3.00 valuations.

PnL Analysis

Defense Metals operates as a pre-revenue developer, so current P&L is dominated by exploration, testwork, and overhead costs. For the year ending March 31, 2025:

- Net Loss: CAD $5.77M (vs. $2.76M prior year), reflecting project advancement and increased spending on PFS, permitting, and engineering.
- No Operating Revenue or EBITDA—entire cash burn funded via equity raises.
- Quarterly Burn Rate: CAD $1.4–1.8M, expected to rise to CAD $2M+ during DFS phase (2026–2027).

Looking forward, commercial P&L depends entirely on project execution. The PFS models annual open-pit production of 31,900 tonnes TREO in concentrate, yielding 5,200 tonnes MREC after stripping cerium and lanthanum. At base-case pricing, modelled operating margins exceed 25%, but cash flow generation is 4–6 years away and requires CAD $1B+ in additional development capital. Ongoing dilution and cost inflation risk remain key, but recent financings and debt elimination materially improve balance sheet flexibility for upcoming feasibility work.

Key Milestones (2026-2030)

- **H1 2026:** Definitive Feasibility Study commencement; metallurgical pilot plant construction; engineering contract awards for mine design, processing plant, and infrastructure
- **H2 2026:** Pilot plant operations and flowsheet optimization; environmental baseline studies completion; indigenous consultation and impact-benefit agreement negotiations
- **H1 2027:** DFS completion and NI 43-101 technical report filing; environmental assessment submission to BC Environmental Assessment Office and Canadian Environmental Assessment Agency
- **H2 2027:** Binding offtake agreement execution with strategic end-user; construction financing roadshow and debt mandates; engineering, procurement, construction (EPC) contractor selection
- **2028:** Environmental assessment approval; provincial mine permit and federal permits secured; financial close on construction financing package; FID (Final Investment Decision) and construction commencement
- **2029:** Plant construction and commissioning; mine development and pre-production stripping; recruitment and training of operations workforce
- **2030:** Commercial production commencement; ramp-up to nameplate capacity of 5,200 tpa MREC; first product shipments to offtake partners

Risks

Commodity Price Volatility Risk (HIGH): NdPr oxide prices are highly cyclical and a sustained 20% decline from current levels would render the project sub-economic at modeled cost structure, jeopardizing financing and potentially forcing asset write-downs. Historical NdPr pricing has ranged from $30/kg (2016 trough) to $180/kg (2022 peak), with volatility driven by Chinese export policy, EV adoption rates, and global macroeconomic conditions. Downside scenarios with $40-50/kg NdPr pricing would reduce after-tax NPV to $100-300M and risk-adjusted equity value to $0.20-0.40 per share.

Execution Risk - Project Development (HIGH): Defense Metals faces multi-year development timeline with numerous binary execution gates, each carrying failure risk: (1) DFS may identify higher capex, lower recovery, or adverse metallurgy requiring design changes (15-20% probability); (2) pilot plant testing may reveal flowsheet issues necessitating additional optimization (10% probability); (3) EPC contractors may deliver cost estimates 20-30% above PFS assumptions due to inflation (30% probability); (4) construction delays or cost overruns could extend timeline by 12-24 months and increase capex by $200-400M (25% probability). Any of these outcomes would materially impair economics and potentially render the project unfinanceable, resulting in significant share price declines or corporate restructuring.

Financing Risk (VERY HIGH): The company requires approximately CAD $1.0-1.2 billion in total financing over 4-6 years, of which only US $250M is preliminarily committed via EDC Letter of Interest. Failure to secure construction debt on favorable terms (target: 60% debt at <7% interest) would force highly dilutive equity raises, potentially 5-10x current share count, reducing per-share economics to $0.20-0.40 even in success scenarios. Strategic investment or offtake prepayments are critical to limit dilution, but negotiations may require accepting unfavorable commercial terms (e.g., below-market pricing, take-or-pay obligations, equity warrants). Project finance markets for junior miners tightened significantly in 2024-2025, with rare earth projects facing additional skepticism due to commodity price volatility and Chinese competitive dynamics.

Permitting & Regulatory Risk (HIGH): British Columbia mine approval timelines historically range 3-7 years, with environmental assessment processes subject to indigenous consultation requirements, public comment periods, and political considerations. Recent BC projects have faced delays due to: (1) indigenous opposition over land rights or environmental concerns (e.g., Coastal GasLink faced 2-3 year delays); (2) environmental NGO challenges on water usage, wildlife habitat, or climate impacts; (3) regulatory changes imposing stricter tailings management or emissions standards. Any permitting delay beyond 2028 would extend Defense Metals' cash burn period, requiring additional dilutive financings and potentially causing strategic partners or lenders to withdraw. The company has secured McLeod Lake Indian Band support via Joint Co-Design Agreement, but other First Nations in the region may assert consultation rights, and provincial/federal governments could impose additional conditions or deny permits.

Shareholder Dilution Risk (VERY HIGH): Defense Metals will require CAD $200-300 million in equity capital before securing project financing, implying 50-100% dilution over the next 2-3 years at current market capitalization of $50-60M. At CAD $0.40-0.50 average raise pricing, this represents 400-700 million new shares, increasing fully diluted count from 110M to 500-800M before construction financing. During construction financing, an additional CAD $500-700M equity component could add 700-1,000M shares if raised at $0.60-0.80 levels, resulting in total share count of 1.0-1.5 billion at first production. While successful project execution could support $2-4/share valuations at production, this implies enterprise values of $2-6B—requiring the company to become a mid-tier rare earth producer with sustained high margins. Investors must underwrite significant dilution risk, with per-share returns dependent on securing strategic investment at premiums to current trading levels.

Offtake Execution Risk (MEDIUM-HIGH): Defense Metals has executed a non-binding MOU for rare earth supply but has not disclosed: (1) counterparty identity; (2) contract volumes or duration; (3) pricing mechanisms; (4) advance payment terms; or (5) binding commitments. Failure to convert the MOU into a binding offtake agreement with meaningful prepayments ($50-100M) would materially increase financing difficulty and potentially require accepting unfavorable commercial terms from strategic investors. Rare earth offtake markets are dominated by Chinese processors and permanent magnet manufacturers who may be unwilling to pay premiums for Western material or may demand pricing linked to Chinese spot markets (negating jurisdiction premium). Western end-users (automotive OEMs, defense contractors) are increasingly interested in secure supply but may lack experience with long-term rare earth procurement or may demand onerous contract terms.

Geopolitical & Market Demand Risk (MEDIUM): Defense Metals' investment thesis depends on sustained Western demand for secure rare earth supply and willingness to pay premiums over Chinese material. However: (1) Chinese government could implement predatory pricing (subsidized rare earth exports at $40-50/kg NdPr) to undermine non-Chinese projects; (2) slower-than-expected EV adoption due to charging infrastructure gaps, battery cost inflation, or consumer preference shifts could reduce magnet rare earth demand growth from 8-10% CAGR to 3-5%; (3) technological substitution (e.g., ferrite magnets, rare earth-free motors) could reduce NdPr intensity per vehicle; (4) geopolitical tensions could ease, reducing Western governments' urgency to diversify supply chains and eliminating premium pricing.

Potential Catalysts (12-18 Months)

DFS Commencement (Q1-Q2 2026): Formal announcement of DFS start with engineering firm selection would validate project advancement and potentially support share price re-rating toward $0.60-0.70 levels.

Binding Offtake Agreement: Execution of commercial supply agreement with automotive OEM, permanent magnet manufacturer, or defense contractor with advance payments of $50-100M would materially de-risk financing and support $0.80-1.00 share price.

Government Funding Award: Receipt of CAD $50-150M grant or low-interest loan from Canada Critical Minerals Infrastructure Fund or U.S. Defense Production Act Title III would reduce equity financing requirements and limit dilution.

NdPr Price Recovery: Sustained recovery in NdPr oxide pricing to $75-100/kg (base case PFS assumptions) driven by EV demand growth or Chinese export restrictions would improve project economics and support higher probability-adjusted valuations.

Permitting Progress: Environmental assessment approval or early-stage permit issuance would reduce timeline risk and support multiple expansion toward development-stage peers.

Conclusion

Defense Metals offers high leverage to rare earths supply chains reshoring, anchored by a robust PFS for a uniquely high-grade, NdPr-focused project in a top-tier jurisdiction. The company's recent financing and debt elimination build a stronger base for upcoming definitive feasibility and permitting. However, material risks, including a multi-year timeline, multi-hundred-million-dollar financing needs, potential shareholder dilution, and exposure to volatile magnet rare earth pricing remain. Successful execution on engineering, permitting, and strategic partnerships could unlock substantial upside. For risk-tolerant investors, Defense Metals represents a high-risk, high-reward play on the western rare earth value chain, but its investment case is dependent on both rare earth market dynamics and disciplined project delivery.